2025 IEEE 30th Pacific Rim International Symposium on Dependable Computing (PRDC) | 979-8-3315-4513-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/PRDC67299.2025.00024

CLGuard: A Context-Aware Suppression Framework for Resilient Driving Control

Yeongju Seong, Hyuk Lim, and Seunghyun Yoon
Dept. of Energy Engineering, Korea Institute of Energy Technology (KENTECH), Naju, Republic of Korea
{yj2023, hlim, syoon}@kentech.ac.kr

***Abstract*—Autonomous vehicles are vulnerable to diverse faults—including CAN injection, controller errors, and AI misjudgments—that can trigger unsafe actions such as sudden acceleration or erratic steering. We propose CLGuard, a context-aware suppression framework that semantically validates control commands using a pre-trained vision-language model (VLM) and a temporal context buffer. At its core, CLGuard uses a contrasting language goal (CLG) formulation to measure the semantic alignment between a vehicle's action and the driving scene. When a control command is found to be inconsistent with the expected behavior in context, CLGuard suppresses it in real time. Unlike signal-level detectors, CLGuard is fault-agnostic and operates as a non-intrusive wrapper. Experiments in CARLA show that CLGuard improves safety and robustness across diverse control fault scenarios.**

***Index Terms*—Autonomous driving, vision-language models, context-aware suppression, semantic anomaly detection**

I. Introduction

Autonomous driving systems are still exposed to a wide range of faults, including CAN injection attacks, actuator glitches, controller errors, and AI misjudgments. These faults can lead to unsafe behaviors such as sudden acceleration or unexpected steering, even when low-level signals themselves appear normal. The challenge is that many of these faults manifest not as obvious signal anomalies but as control actions that are inconsistent with the semantic expectations of the surrounding driving scene.

Previous approaches to anomaly detection in autonomous driving have mainly focused on narrow categories of failures. For example, CAN intrusion detectors employ statistical models or recurrent neural networks to detect abnormal signal patterns [1], yet they do not evaluate whether the resulting actions are behaviorally appropriate. Similarly, recent works with Vision-Language Models (VLMs) such as CLIP have been applied to identify visual anomalies [2], while others use VLM feedback to influence planning or reinforcement learning rewards [3]. However, these approaches are limited to perception or offline reasoning and do not provide a mechanism for suppressing unsafe control actions in real time, regardless of the fault source.

In this work we introduce **CLGuard**, a lightweight context-aware framework that validates control commands before execution and intervenes when they are unsafe. CLGuard assesses the consistency between a vehicle's observations and its issued actions using a pre-trained vision–language model guided by Contrasting Language Goals (CLG), while a temporal context buffer incorporates recent sensor states and control history to stabilize decisions. When an action is judged to be semantically inconsistent with the driving scene, CLGuard suppresses the command and replaces it with a safe fallback. Our contributions are threefold. We present semantic and fault-agnostic runtime suppression for driving control through CLG-based scene–action alignment, we design a deployable wrapper that requires no retraining of the existing controller, and we validate its effectiveness in simulation through diverse injected fault scenarios. This design makes CLGuard a practical wrapper that can be integrated with existing autonomy stacks without retraining or intrusive modifications.

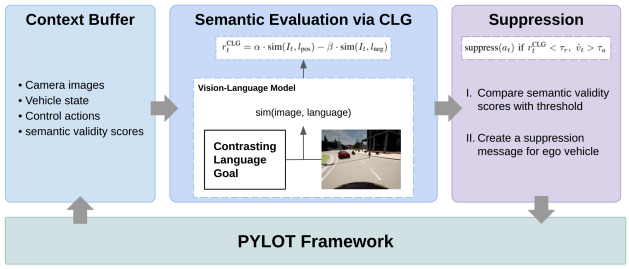

Fig. 1. Architecture of the proposed CLGuard Framework.

II. Proposed Method

CLGuard detects and suppresses unsafe control actions by combining real-time sensor input, vehicle state, and control history with semantic reasoning from a VLM. Rather than modeling specific failure types, it evaluates whether each issued command is semantically consistent with the observed driving scene.

A. Multi-Modal Context Buffer

At each timestep t, we maintain a rolling buffer $\mathcal{C}_t$ that aggregates recent multimodal information:

- $I_{t-k:t}$: front-view RGB images,
- $s_{t-k:t}$: vehicle state (velocity, acceleration, yaw),
- $a_{t-k:t}$: recent control actions (throttle, brake, steer),
- $r^{\mathrm{CLG}}_{t-k:t}$: semantic validity scores.

This temporal context allows the framework to judge whether the current action a_t aligns with expected safe behavior.

2473-3105/25/$31.00 ©2025 IEEE
DOI 10.1109/PRDC67299.2025.00024

Authorized licensed use limited to: Universiti Malaysia Sabah. Downloaded on April 09,2026 at 04:17:18 UTC from IEEE Xplore. Restrictions apply.

B. Semantic Evaluation via CLG

To measure scene–action consistency, we adopt a CLG formulation. CLG consists of positive and negative goals. The language goals are selected from a predefined set of sentences, reflecting vehicle state within a temporal context buffer. Given a positive goal (e.g., "the car is driving safely at an appropriate speed") and a negative goal (e.g., "the car is accelerating dangerously into traffic"), we compute

$$r_t^{\mathrm{CLG}} = \alpha \,\mathrm{sim}(I_t, l_{\mathrm{pos}}) - \beta \,\mathrm{sim}(I_t, l_{\mathrm{neg}}), \tag{1}$$

where $\mathrm{sim}(\cdot)$ is cosine similarity in the CLIP embedding space and $\alpha+\beta = 1$. A higher r_t^{CLG} implies stronger alignment with safe intent, while a lower value signals potential anomalies. Although the CLG score is computed on the current frame I_t, the temporal buffer $\mathcal{C}_t$ smooths score variations over time to prevent false suppressions caused by transient visual noise or momentary control fluctuations.

C. Suppression Policy

CLGuard issues a suppression whenever the semantic score and vehicle dynamics jointly indicate risk:

$$\mathrm{suppress}(a_t) \;\; \text{if} \;\; r_t^{\mathrm{CLG}} < \tau_r \;\wedge\; \big(|\dot{v}_t| > \tau_{\mathrm{acc}} \;\vee\; |\dot{\psi}_t| > \tau_{\mathrm{yaw}}\big), \tag{2}$$

where $\dot{v}_t$ is the longitudinal acceleration and $\dot{\psi}_t$ is the yaw rate. This policy indicates a risky condition when the semantic score r_t^{CLG} falls below the safety threshold τ_r and either the vehicle's acceleration magnitude or yaw-rate exceeds its respective limit. In such cases, the unsafe control command a_t is overridden by a safe fallback (e.g., reduced throttle or steering neutralization). This multimodal criterion enables real-time, fault-agnostic suppression of both longitudinal and lateral anomalies without modifying the underlying controller.

III. Evaluation

A. Experimental Setup and Fault Injection

We evaluate CLGuard in the CARLA 0.9.10.1 simulator [4] using Town03 and integrate it with the PYLOT framework [5] for vehicle control. The setup includes a front-facing RGB camera (224×224 resolution) and access to vehicle state such as velocity and acceleration. For semantic scoring we use the OpenCLIP ViT-B/32 model pretrained on LAION-2B, running in real time on a single GPU with an average per-frame latency of 27 ms. To test fault resilience, we inject two representative control anomalies at random intervals: (i) CAN injection, where steering commands are overwritten with random values for 3 s, and (ii) actuator glitch, where brake commands are ignored while throttle is fixed at maximum. Each anomaly is tested in 30 episodes of 40–50 s, covering straight roads and intersections.

B. Metrics

We evaluate CLGuard using three metrics: (i) collision rate, defined as the proportion of episodes in which a collision occurs, (ii) false suppression rate, measured as the frequency of incorrect suppression when no collision is imminent, and (iii) latency, measured as the average end-to-end time from receiving a sensor input to issuing a suppression decision.

C. Results

CLGuard consistently reduces collisions compared to the attacked baseline. This baseline performs its scenario course, omitting any deceleration or suppression actions under attack. Quantitative results in Table I confirm that CLGuard lowers the collision rate while maintaining low latency, demonstrating that semantic suppression can be applied in real time. Our method robustly distinguishes the unsafe control action related to collision risk, without overly interfering with safe control actions.

TABLE I
Suppression and Safety Performance

Method	Coll. Rate	False Supp.	Latency (ms)
Baseline (No Filter)	45.8%	–	–
CLGuard	**6.7%**	**7.9%**	27.5

Overall, suppression decisions follow the trend of the CLG score, indicating that the framework captures meaningful scene–action inconsistencies and generalizes across different fault types without requiring explicit signal-level models. While our evaluation focuses on two representative anomalies, the framework is extensible to a broader range of control and perception faults.

IV. Conclusion

This paper presents CLGuard, a lightweight context-aware framework that semantically validates and suppresses unsafe control actions in autonomous driving. By combining a vision–language model with a temporal context buffer, CLGuard detects scene–action mismatches through CLG-based reasoning and intervenes in real time. Future work will extend the approach to multi-agent settings, explore formal semantic safety guarantees, and move toward real-world validation.

Acknowledgments

This work was partially supported by the IITP grant funded by the Korea government (MSIT) (No. 2021-0-00379), and the Korea Evaluation Institute of Industrial Technology (Alchemist Project, RS-2025-02634277).

References

[1] S. Jeong, S. Lee, H. Lee, and H. K. Kim, "X-CANIDS: Signal-aware explainable intrusion detection system for controller area network-based in-vehicle network," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 3, pp. 3230–3246, 2023.

[2] Q. Zhou, G. Pang, Y. Tian, S. He, and J. Chen, "AnomalyCLIP: Object-agnostic prompt learning for zero-shot anomaly detection," in *International Conference on Learning Representations (ICLR)*, 2024.

[3] Z. Huang, Z. Sheng, Y. Qu, J. You, and S. Chen, "VLM-RL: A unified vision language models and reinforcement learning framework for safe autonomous driving," *arXiv preprint arXiv:2412.15544*, 2024.

[4] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun, "CARLA: An open urban driving simulator," in *Conference on robot learning*. PMLR, 2017, pp. 1–16.

[5] S. Choudhury *et al.*, "PYLOT: An autonomous vehicle platform," https://github.com/erdos-project/pylot, accessed: Sep. 2025.

Authorized licensed use limited to: Universiti Malaysia Sabah. Downloaded on April 09,2026 at 04:17:18 UTC from IEEE Xplore. Restrictions apply.